



TEAM MARKETDEBATERS · HACKATHON SUBMISSION

MarketDebater Swarm

A debate-based AI investment committee for US-listed stocks

THE DIFFERENTIATOR

Hallucinated numbers don't reach the verdict — a deterministic Python verifier catches them first.

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LIVE DEMO

web-production-f16ae.up.railway.app

THE PROBLEM

Retail investors get one-sided AI advice

Chatbot consensus

ChatGPT, Gemini, Copilot all present a single confident opinion. No dissent, no devil's advocate, no risk frame.

Hallucinated stats

Numbers in LLM-generated investment write-ups are wrong ~10–20% of the time and the user has no way to know which.

Information overload

Prices, news, 10-Ks, 10-Qs, 8-Ks, XBRL fundamentals — no single workflow grounds an opinion against all three at once.

→ **Investors need visible disagreement and verifiable claims, not another confident voice.**

THE SOLUTION

A debate, fact-checked by code

- Three rival persona LLMs argue each stock — PermaBull, PermaBear, Risk Manager.
- All three argue from one immutable snapshot of prices, news, and SEC filings.
- A deterministic Python verifier re-checks every numeric claim against the snapshot.
- A separate Chair LLM weights only verified arguments and issues a transparent verdict.
- Verdicts persist to a daily ranked leaderboard — top buys and top shorts in one view.

VERDICT ■ AAPL · 1 ROUND

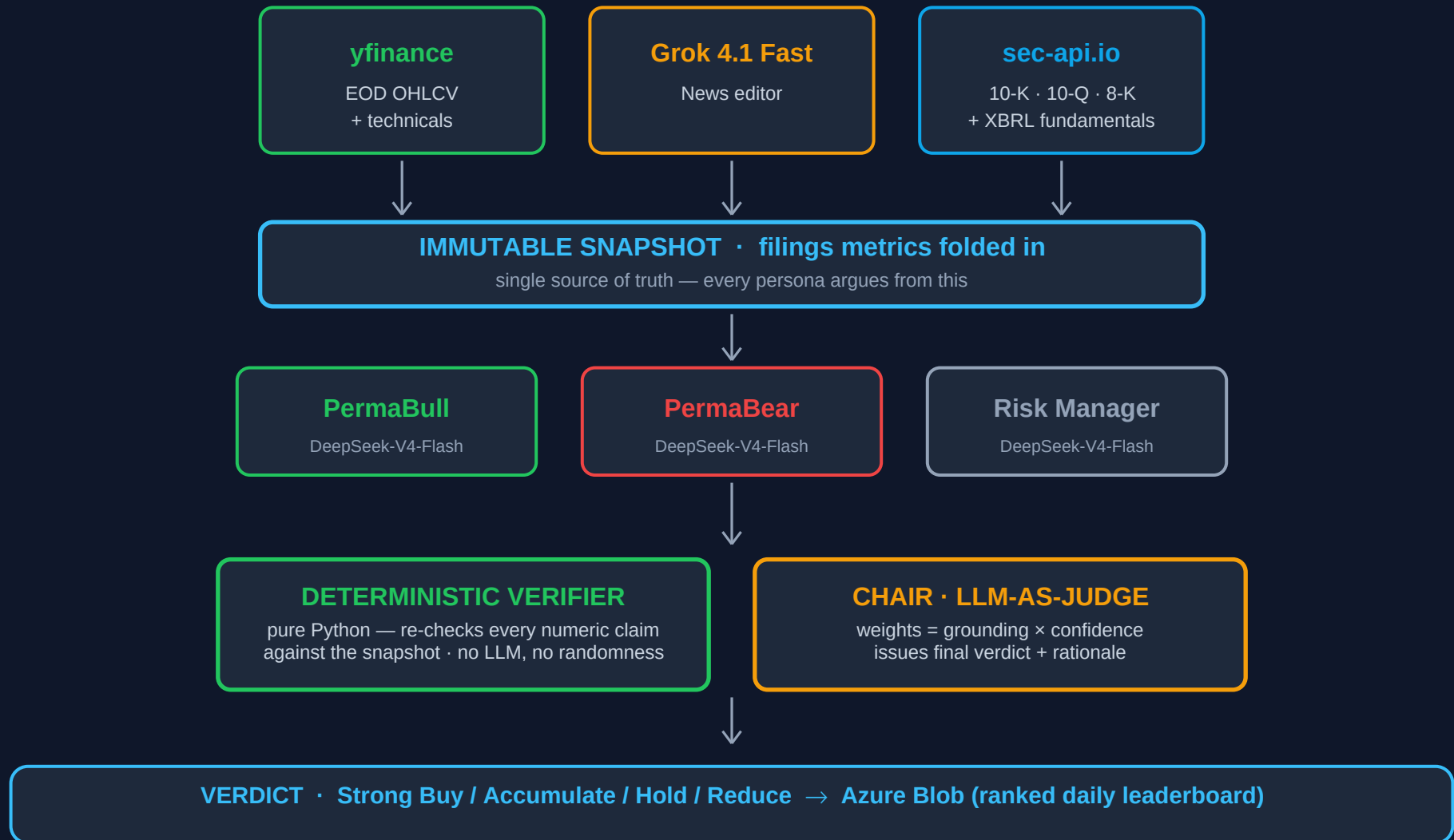
ACCUMULATE



Net bull score = bull weight – bear weight = +0.47

ARCHITECTURE

Five tiers, one immutable snapshot



THE DIFFERENTIATOR

Hallucinated numbers don't reach the verdict

✗ HALLUCINATED CLAIM

PermaBull says:

"AAPL's revenue=\$999B beats every peer — clear buy."

```
verify_grounding(claim, snapshot):  
    truth = snapshot['revenue'] # 395.76B  
    claimed = 999_000_000_000  
    if abs(claimed - truth)/truth > 0.05:  
        reject_as_hallucination()
```

→ PermaBull's weight slashed by the Chair

✓ VERIFIED CLAIM

PermaBull says:

"AAPL's RSI=42 leaves headroom for a move up."

```
verify_grounding(claim, snapshot):  
    truth = snapshot['rsi_14'] # 42.50  
    claimed = 42.3  
    if abs(claimed - truth)/truth < 0.05:  
        accept(grounding=1.0)
```

→ Full weight carried into the verdict

If grounding were an LLM call, this whole project would be theatre.

It's pure Python with regex, value lookups, and arithmetic recomputation. Unit tests are first-class.

AI INTEGRATION

Three LLM roles, one Azure key

PERSONAS

DeepSeek-V4-Flash

(DeepSeek-R1 wired, off by default)

Three rival arguments per ticker per round.

Same model, 3 system prompts.

Structured JSON output:

```
{stance, points, confidence}
```

Each point ships a cited_metric=name=value.

Round 3 must address top opponent point.

CHAIR (JUDGE)

DeepSeek-V4-Flash

(separate role from personas)

Weights only verified claims; issues verdict.

Reads grounding × confidence

Picks Strong Buy / Accumulate / Hold / Reduce.

Emits a transparent rationale.

Different prompt = different blind spots than the personas it judges.

NEWS EDITOR

Grok 4.1 Fast Reasoning

(via Azure AI Foundry catalog)

Filters and summarizes RSS to top items.

Pulls 20 RSS headlines per ticker.

Selects 8 most material; tight summaries.

Same Azure key, no separate xAI account.

Disable via GROK_MODEL="" to fall back to raw RSS without LLM editing.

All three reachable through Azure AI Foundry's OpenAI-compatible /openai/v1 endpoint · one key, three model deployments

DEMO — SINGLE DEBATE

Verdict page · grounded debate transcript

■ Educational only — not financial advice

Single Debate

Daily Suggestions

Apple Inc.

HOLD

\$307.34

+1.42%

PermaBull

Grounding	0.85
Confidence	0.78
Weight	0.660

PermaBear

Grounding	0.62
Confidence	0.55
Weight	0.340

Risk Manager

Grounding	0.90
Confidence	0.60
Weight	0.540

Persona Deep-dive ■

1 round · each persona's claims, citations & sources

ROUND 1 · CASE

■ Trading at \$307.34, below 52-week high of \$321 — room to run.

cited_metric=week52_high=321.0

DEMO — DAILY SUGGESTIONS

Leaderboard · top buys and top shorts in one view

Single Debate **Daily Suggestions**

Run a fresh scan

Trigger a synchronous scan; refreshes the ranking below.

AAPL, MSFT, GOOGL, NVDA, AMZN

Run scan

#	TICKER	COMPANY	VERDICT	NET BULL
1	GOOGL	Alphabet Inc.	HOLD	+0.073
2	AAPL	Apple Inc.	HOLD	+0.022
3	AMZN	Amazon Inc.	HOLD	0.000
4	MSFT	Microsoft Corp.	HOLD	-0.171
5	NVDA	NVIDIA Corp.	REDUCE	-0.402

TECH STACK & DEPLOYMENT

Microsoft-first stack, deploy-anywhere shape

LLM BRAIN

- Azure AI Foundry
- /openai/v1 (OpenAI-compatible)
- DeepSeek-V4-Flash · DeepSeek-R1
- Grok 4.1 Fast Reasoning
- Single key, three deployments

FRONTEND

- Django 6.0 + WhiteNoise
- gunicorn (2 workers · 30-min timeout)
- Streamlit (collaborator-owned)
- Same data contract for both UIs

DATA SOURCES

- yfinance — EOD OHLCV + technicals
- sec-api.io — 10-K · 10-Q · 8-K + XBRL
- Google News RSS → Grok edited
- Watchlist or Google Sheet universe

PERSISTENCE

- Azure Blob Storage
- verdicts/<date>/<ticker>.json
- verdicts/<date>/index.json (ranked)
- LOCAL_RESULTS_DIR fallback for dev

SEARCH / RAG

- Azure AI Search (BM25)
- Indexes news + filings together
- source_type filter at query time
- Graceful no-op when unconfigured

HOSTING & CI

- Railway (~\$5–10 / month)
- Auto-deploy from main
- GitHub Actions: on-demand scan
- Procfile + WhiteNoise = no nginx
- Live: web-production-f16ae.up.railway.app

WHAT'S NEXT

Future versions yet to come

What ships next, beyond the hackathon scope — bigger universe, deeper data, real production posture.

01

Wider US universe

- Nasdaq 100, S&P 500, Russell 3000.
- 4.6K-ticker sheet already wired —
- one config flip + paid sec-api tier.

02

IPO coverage

- Ingest S-1 prospectuses pre-listing.
- Track lockup expiries + dilution.
- Day-one debate when pricing drops.

03

Other markets

- LSE, HKEX, TSE via yfinance suffix.
- Return to NSE / BSE where this began.
- Per-market filings adapter.

04

Earnings calls

- Pull transcripts post-earnings.
- Persona Q&A on management tone.
- Sentiment delta into the snapshot.

05

Backtested accuracy

- Per-persona win rate vs returns.
- Chair auto-learns weighting.
- Confidence calibrated on history.

06

Streaming + mobile

- SSE live transcripts — no 5-min wait.
- Background workers for bulk scans.
- PWA so it works on phones.